

Integrating Physical Activity and VR-Based Mindfulness to Support Stress and Cardiometabolic Health in Breast Cancer Patients: A Feasibility Study

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Background & Problem Statement: Breast cancer (BC) is one of the most prevalent cancers among women (American Cancer Society 2025). Even in survivorship, psychological distress (Tang et al. 2024), physical discomfort, and cardiometabolic risk (Jung et al. 2025) are common. Clinically significant psychological symptoms are reported in up to 50% of patients with BC during the stages of diagnosis, treatment, and rehabilitation, seriously impacting quality of life (QOL). Depression, anxiety, and stress symptoms are particularly high in this patient population (Zhang et al. 2022) and comprehensive care models are proposed to not only support patient outcomes through medical intervention but also by boosting access to psychological support and mental health resources (Singer et al. 2010). The physical symptom burden in patients with breast cancer is equally problematic. Patients evaluated by the Distress Thermometer and Problem List (DT&PL) report high levels of fatigue, sleep disturbance, pain, neurological compromise, as well as other organ and systemic impact (McFarland et al. 2018). These outcomes are often consequences of interventional treatments like surgery, radiation, chemotherapy, and hormone modifying pharmacologics, and they strongly correlate to psychological distress and QOL scores (McFarland et al. 2018). Medical interventions for breast cancer also impact cardiometabolic health, including higher risks of hypertension and type 2 diabetes. Among women with early stage breast cancer, cardiometabolic risk factors were significantly associated with an increased development of heart failure (Mavrogeni et al. 2019) and with cardiovascular and all-cause mortality (Simon et al. 2018). Finally, treatments for BC, such as chemotherapy and hormone therapy can impact body composition, metabolism, inflammation, and physical endurance (Pin et al. 2018), however, attenuation of these outcomes may be possible through exercise interventions and stress management (Fig. 1).



Figure 1. Approaches to breast cancer recovery

Physical activity [PA] (Carayol et al. 2019), including exercise before, during, and after chemotherapy in high-risk breast cancer patients was associated with a significantly reduced hazard of recurrence and mortality (Cannioto, et al. 2021), however studies utilizing activity trackers have been equivocal in terms of improvement of PA levels. (Dreher, et al, 2019). Mindfulness-based interventions (Kenne et al. 2017; Liu et al. 2022) have been found to be beneficial in improving outcomes of physical and psychological distress. Mindfulness-based therapy introduced to patients undergoing breast cancer treatment was effective at improving anxiety, depression, fatigue, spiritual well being, fear of cancer reoccurrence (FCR), and quality of life (QOL) (Lengacher et al. 2021, Zhang et al. 2022). Virtual reality (VR) is a promising and immersive technology that has the potential to improve the engagement and reach of mindfulness interventions (Arpaia et al. 2022), particularly in populations that are undergoing intensive treatment (such as chemotherapy infusion) or long term

recovery. Immersive VR technology offers a unique platform that can address the obstacles often associated with traditional mindfulness methods such as accessibility, convenience, and engagement. **However, there has been limited exploration in the field to test the combined effects of patient-centered exercise with activity trackers and mindfulness, especially when delivered through innovative platforms such as immersive virtual reality (VR).**

Specific Aims:

Aim 1: Assess the feasibility of implementing a 12-week combined PA (with FitBit trackers) and VR-based mindfulness intervention in patients diagnosed with BC. *Feasibility will be evaluated through recruitment and retention rates, adherence, and participant satisfaction.*

Aim 2: Evaluate preliminary effects of the intervention on perceived stress and cardiometabolic health. *Outcomes will include changes in biomarkers (e.g., cortisol, C-reactive protein [CRP]), self-reported stress, and cardiometabolic indicators (e.g., blood pressure, RHR, BMI).*

Hypotheses: H1: the combined intervention will be feasible and acceptable to participants. H2: the combined intervention will lead to improvements in perceived stress and cardiometabolic health compared to baseline measures.

Methodical Approach and Timeline:

Experimental Approach: We propose a 12-week feasibility study involving patients recently diagnosed with BC and initiating chemotherapy, to explore the impact of PA with FitBit trackers and VR mindfulness training on perceived stress and cardiometabolic health. Participants: A total of 10 participants diagnosed with BC and initiating chemotherapy treatment, will be recruited through physicians at the Stony Brook Cancer Center. No subpopulations will be excluded on the basis of sex, race, or ethnicity. Inclusion criteria: 1) 25-75 years of age, 2) diagnosed/biopsy-proven early-stage breast cancer and initiating chemotherapy (any IV infusion form) with intent to cure, 3) willing to use FitBit trackers and VR equipment and to participate in an exercise regimen. Exclusion criteria: 1) less than 25 or older than 75 years old, 2) incurable or metastatic disease and received chemotherapy previously, 3) unwilling to participate with VR tool and exercise protocol, and 4) underlying cardiac or pulmonary diseases sufficient to make exercise impractical or unsafe. Participants will be enrolled on a rolling basis throughout the year (fall 2025 and spring 2026) and study timepoints (below) will align with initiation of each subject's medical treatment. A protocol will be submitted to the SBU Institutional Review Board (IRB) for approval prior to the recruitment of subjects and initiation of any study protocols. Subject participation will be incentivized by providing participants a FitBit Inspire 3 and exercise equipment package at the start of the intervention, as well as a gift card for data collection completion.

Interventions: Participants will receive a patient-centered PA and VR mindfulness intervention along with a wearable tracking device (FitBit: Inspire 3) to monitor and record key health metrics. In addition to the home exercise component, participants will be provided with a VR unit and be asked to complete a minimum of three 20-minute mindfulness sessions per week for 12 weeks using the VR headset. The headset will be returned following the study. Prior to enrollment, all participants will undergo informed consent process in accordance with IRB guidelines. They will also receive training on the use of the FitBit, VR hardware, and receive exercise-related support through study

Primary Outcomes:

- Statistics:** Analyses will be conducted using SPSS v29. Descriptive statistics will be completed to evaluate both the T01 and T02 outcomes. Paired t-tests (using SNK post hoc) will be run to determine if significant differences ($P < 0.05$) exist in primary outcomes. It is anticipated that statistical power will be low given the low sample size for this feasibility study, but variability and effect size in data will guide sample size determination for a larger RCT study design.

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Potential Study Impact: The long-term goal of this project is to develop an integrated, scalable and patient-centered intervention using technology such as PA trackers and VR units, that can enhance quality of life and long-term health outcomes among breast cancer survivors—and potentially extend to individuals surviving other cancer types. This aim is closely aligned with the NIH mission to support research that improves health, prolongs life, and reduces the burden of illness and disability.

Roles and responsibilities of PI and Co-Investigators: This study includes an interdisciplinary team of investigators at Stony Brook University. The team structure provides expertise in the assessment and interpretation of change to medical, physical, and mental health outcomes in the subject population. Kelly Warren (physiologist, School of Health Professions) will serve as PI and be responsible for study oversight, including submission of IRB and future grant applications. She will organize the team to execute the 12-week intervention, and lead data collection, analysis, and interpretation. Sharon Martino (physical therapist, School of Health Professions) will guide the 12-week exercise program and participate in data collection of anthropometrics and exercise endurance. Jules Cohen (oncologist, RSOM Department of Medicine) will recruit and screen participants, order and interpret blood tests and results, as well as monitor for adverse occurrences during the protocol. Adam Gonzales (psychologist, Vice Chair of Behavioral Health, Department of Psychiatry & Behavioral Health, RSOM) will provide expertise in mental health screening and outcomes.

Plans for Future Grant Submissions and Dissemination: Findings from this feasibility study will inform the development of a larger, fully powered randomized controlled trial to evaluate the efficacy of a combined physical activity with trackers and virtual reality (VR)-based mindfulness intervention in improving stress and cardiometabolic health outcomes among breast cancer patients. Given the high prevalence of breast cancer among women and the growing population of survivors — many of whom experience persistent psychological distress and elevated cardiometabolic risk following treatment — this line of research is both timely and aligned with national health priorities. Future grant submissions will target both internal and external mechanisms of funding. A TRO will be submitted in September 2025 through the Stony Brook Cancer Center. External opportunities include NIH mechanisms such as the R21 (Exploratory/Developmental Research Grant) to further refine the intervention and assess preliminary efficacy, as well as the National Cancer Institute (NCI) and the National Center for Complementary and Integrative Health (NCCIH), particularly under initiatives focused on behavioral medicine, digital health, and integrative oncology. These submissions will take place in Spring/Summer 2026 when the majority of data collection and analysis is completed. Dissemination efforts will include the submission for conference presentation and development of a peer-reviewed publication.

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